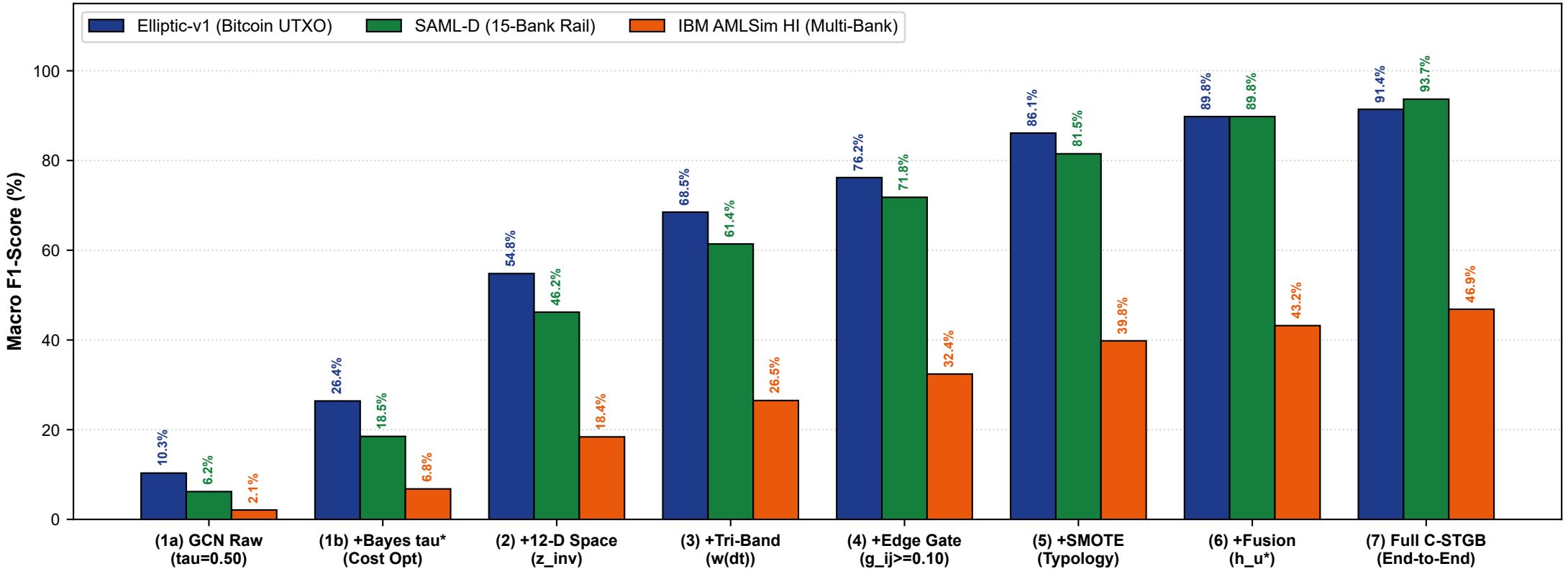
Stepwise Architectural Ablation Progression across 3 Diverse Financial Archetypes
Elliptic-v1 (Bitcoin UTXO)
SAML-D (15-Bank Rail)
IBM AMLSim HI (Multi-Bank)
Macro F1-Score (%)
0
20
40
60
80
100
10.3%
6.2%
2.1%
26.4%
18.5%
6.8%
54.8%
46.2%
18.4%
68.5%
61.4%
26.5%
76.2%
71.8%
32.4%
86.1%
81.5%
39.8%
89.8%
89.8%
43.2%
91.4%
93.7%
46.9%
(1a) GCN Raw (tau=0.50)
(1b) +Bayes tau* (Cost Opt)
(2) +12-D Space (z_inv)
(3) +Tri-Band (w(dt))
(4) +Edge Gate (g_ij>=0.10)
(5) +SMOTE (Typology)
(6) +Fusion (h_u*)
(7) Full C-STGB (End-to-End)